From List-Decodability to Proximity Gaps

Yiwen Gao[1], Dongliang Cai[1], Yang Xu[1], and Haibin Kan[1]

Fudan University, Shanghai, China
ywgao21@m.fudan.edu.cn, dlcai22@m.fudan.edu.cn, xuyyang@fudan.edu.cn, hbkan@fudan.edu.cn

Abstract. Proximity testing for linear codes is a fundamental problem in coding theory with critical applications in cryptographic protocols, blockchain, and distributed storage systems. This work addresses the proximity gaps for linear codes, a crucial aspect for efficiently verifying whether a batch of codewords is close to a given code. We present a general framework for deriving proximity gaps from the list-decodability properties of the underlying linear code.

Our main result shows that if a code $C \subseteq \mathbb{F}_q^n$ is (p, L)-list-decodable, then the probability that a random combination of a batch of t codewords containing a δ-far codeword (for $\delta \leq 1 - \sqrt{1 - p + \varepsilon}$) remains δ-far from C is bounded by $O(\frac{tL^2pn}{q} + \frac{t}{\varepsilon q})$. This result also establishes a form of (mutual) correlated agreement for linear codes, which can be used to strengthen soundness analyses in protocols that rely on proximity testing, thereby reducing query complexity and enabling practical instantiations over smaller finite fields. In particular, we apply our main result to randomly punctured Reed–Solomon codes and folded Reed–Solomon codes—both of which are known to achieve list-decodability up to capacity—and derive linear proximity gaps for these families under the Johnson bound.

Keywords: Proximity gaps · Linear codes · Mutual correlated agreement.

1 Introduction

A linear code $C \subseteq \mathbb{F}_q^n$ is a subspace of $\mathbb{F}_q^n$, whose elements—referred to as codewords—have length n. Linear codes have a wide range of applications. In particular, many protocols in areas such as blockchain, distributed storage, and cryptography utilize different linear codes as essential building blocks. These protocols assume oracle access to a batch of codewords $\{\pi_0, \ldots, \pi_t\}$, and their soundness requires that all of these codewords are close (in relative Hamming distance) to a fixed linear code C. Consequently, it is critical to efficiently identify vectors that are far from the linear code.

The *Proximity Testing* problem of linear codes involves a verifier determining whether a given codeword $\pi \in \mathbb{F}_q^n$ is close to a given linear code C or is far from all the members of C. The verifier has limited query access to π, and an untrusted prover may assist the verifier. We consider this problem under the interactive oracle proofs of proximity (IOPP) model [BCS16] (also called probabilistically checkable interactive proofs of proximity in [RRR16]). This model combines aspects of probabilistically checkable proof (PCPs) and interactive proofs (IPs). The prover sends auxiliary messages, and the verifier has oracle access to both the input π and the prover's messages.

For a batch of codewords $\{\pi_0, \ldots, \pi_t\} \subseteq \mathbb{F}_q^n$, one can implement a protocol for the proximity testing problem on each codeword to ensure that they are all close to C. However, this approach is inefficient when the batch size is large. [RVW13] provides an approach: randomly choose a vector π' in the span of $\{\pi_0, \ldots, \pi_t\}$ (denoted by $\mathrm{span}(\pi_0, \ldots, \pi_t)$) and check if π' is close to C. The soundness proof of this method raises an important question: *If $\exists \pi_i \in \{\pi_0, \ldots, \pi_t\}$ that is far from all the members of C, can we prove π' is far from C with high probability?* In other words, we are asked to provide an upper bound on the probability that a random batch π' that *discloses* the distance. Many previous works have explored this question and provided positive answers. This property is referred to as the *proximity gap* for linear codes, a notion formally introduced by Ben-Sasson, Carmon, Ishai, Kopparty, and Saraf in [Ben+20b]. Furthermore, they provide proximity gaps for Reed-Solomon (RS) codes[RS60] under the Johnson bound[Joh62].

1.1 List decoding

The proximity gap for a linear code is closely related to its list-decodability. For a codeword $c \in C$, let $\pi \in \mathbb{F}_q^n$ be its corrupted version with error in some positions. The *unique decoding* problem aims to recover c from π. The *list decoding* problem is a generalization of the unique decoding problem[Eli57][Woz58]. For a given codeword $\pi \in \mathbb{F}_q^n$ and $p \in [0, 1]$, we want to find all the codewords $c \in C$ that are p-close to π in relative Hamming distance. If for any $\pi \in \mathbb{F}_q^n$, there are at most L codewords that are p-close to π, we say the code C is (p, L)-list decodable. The results of [Ben+20b] are based on two different decoding algorithms for RS codes: the Berlekamp-Welch unique decoding algorithm, and the Guruswami-Sudan list decoding algorithm [GS98].

The list-decodability of a linear code has been studied for a long time. When the distance p is under the Johnson bound, any code with a large minimal distance has non-trivial list-decodability[Gur+07]. Especially, for RS codes, Guruswami and Sudan[GS98] provide an efficient list-decoding algorithm up to the Johnson bound. The list-decodability of codes beyond the Johnson bound is much harder.

Over the past decade, an exciting long line of works [RW14; ST20; Guo+24; FKS22; GST22; BGM23; GZ23; AGL24] show that RS codes over random evaluation domains are, with high probability (over the choice of evaluation points), list-decodable much beyond the Johnson bound. In particular, Shangguan and Tamo [ST20] introduced the *generalized Singleton bound* and conjectured that generic Reed-Solomon codes can achieve list-decoding capacity. Later, a breakthrough of Brakensiek, Gopi, and Makam [BGM23] resolved this conjecture and showed that Reed-Solomon codes are list-decodable up to list-decoding capacity with exponential-sized alphabets. The following exciting works by Guo and Zhang [GZ23] and Alrabiah, Guruswami, and Li [AGL24] improved the alphabet size to linear, i.e., with high probability, a randomly punctured RS code with rate R is $(1-R-\varepsilon, O(1/\varepsilon))$ list-decodable. Besides randomly punctured RS codes, Chen and Zhang [CZ24] showed that folded RS codes and univariate multiplicity codes with rate R are $(1-R-\varepsilon, O(\frac{1}{\varepsilon}))$ list-decodable, implying the first explicit codes that are list-decodable up to list-decoding capacity with polynomial-sized alphabets.

1.2 Our results

In this work, we present a general framework for deriving proximity gaps from the list-decodability properties of the underlying linear codes. Our method does not require a concrete list-decoding algorithm. More precisely, we establish the following results for different types of combination structures:

Theorem 1 (Informal: Plain Combination). *Let $C \subseteq \mathbb{F}_q^n$ be a (p, L) list-decodable linear code, $\varepsilon > 0, \delta \leq 1 - \sqrt{1-p+\varepsilon}$. Let $\pi_0, \ldots, \pi_t \in \mathbb{F}_q^n$ be a batch of codewords. Suppose $\exists \pi_i \in \{\pi_0, \ldots, \pi_t\}$, such that $\delta(\pi_i, C) \geq \delta$, where $\delta(\pi_i, C)$ denotes the relative Hamming distance. We have*

$$\Pr_{\alpha_1,\ldots,\alpha_t \in \mathbb{F}_q}(\delta(\pi_0 + \alpha_1\pi_1 + \ldots + \alpha_t\pi_t, C) < \delta) < \frac{t}{q} \cdot (L^2 pn + \frac{1}{\varepsilon}). \quad (1)$$

In addition to the plain linear combination structure, which is useful in batching-based proximity testing, we extend our analysis to a tensor combination structure. This type of structure arises in certain proximity testing protocols that employ folding as a core technique, notably in FRI[Ben+18].

Theorem 2 (Informal: Tensor Combination). *Let $C \subseteq \mathbb{F}_q^n$ be a (p, L) list-decodable linear code, $\varepsilon > 0, \delta \leq 1 - \sqrt{1-p+\varepsilon}$. Let $\pi_1, \ldots, \pi_{2^t} \in \mathbb{F}_q^n$ be a batch of codewords. Suppose $\exists \pi_i \in \{\pi_1, \ldots, \pi_{2^t}\}$, such that $\delta(\pi_i, C) \geq \delta$, where $\delta(\pi_i, C)$ denotes the relative Hamming distance. We have*

$$\Pr_{\alpha_1,\ldots,\alpha_t \in \mathbb{F}_q}(\delta\left(\langle \pi_1, \ldots, \pi_{2^t}\rangle \cdot \otimes_{i=1}^t \langle 1, \alpha_i\rangle, c\right) < \delta) < \frac{2^t - 1}{q} \cdot (L^2 pn + \frac{1}{\varepsilon}). \quad (2)$$

We show that if one of the inputs in the batch is δ-far from the linear code, then with the probability that is linear in the code length (over the choice of combining points), the combining result is δ-far from the linear code. We refer to this probability as the *bad-combining* probability. The formal versions of these statements, which are stronger, are presented in Corollary 3 and Corollary 4.

Mutual correlated agreement. We utilize the concept of *correlated agreement*, as defined in [Ben+20b]. A batch of codewords $\pi_0, \ldots, \pi_t$ have δ-correlated agreement with C if there exists a sufficiently large subdomain $D \subseteq [n]$ and $c_0, \ldots, c_t \in C$ such that

$$|D| \geq (1-\delta)n \text{ and } \pi_i[D] = c_i[D], 0 \leq i \leq t,$$

where $\pi_i[D]$ denotes the sub-codeword of π_i on D. The definition of correlated agreement is relevant in the context of real-world protocol applications. Notice that even if all of π_i are δ-close to C, they may not have δ-correlated agreement. Our formal theorem supports this stronger notion of agreement.

Our result not only contains the combining points that explicitly disclose the distance, but also accounts for those that have *potential* to disclose the distance. More precisely, for a given $\delta \leq 1 - \sqrt{1 - p + \varepsilon}$, $\pi_0, \ldots, \pi_t$ may have correlated agreement on $[n]$. Let $\{D_1, D_2, \ldots, D_m\}$ denote all the correlated-agree-domains satisfying $|D_i| \geq (1 - \delta)n$. We demonstrate that with high probability, the δ-agree-domains preserved in the combining result $\pi_0 + \alpha_1\pi_1 + \ldots + \alpha_t\pi_t$, i.e., coincide with $\{D_1, D_2, \ldots, D_m\}$. This property is called *mutual correlated agreement* and can be applied to prove the soundness of some recent protocols [Arn+25][Bün+25].

Proximity gaps for two specific RS codes. We apply the main result to randomly punctured RS codes and folded RS codes, which have been proved to be list-decodable up to capacity. We prove that they have linear proximity gaps under the Johnson bound. More precisely, we have:

- **Randomly punctured RS codes.** With high probability (over the choice of evaluation domain), a randomly punctured RS code C is $(1 - R - \varepsilon, O(\frac{1}{\varepsilon}))$ list-decodable [AGL24]. Let R be the rate of C and $\varepsilon > 0, \delta \leq 1 - \sqrt{R + 2\varepsilon}$. For a batch of codewords $\pi_0, \ldots, \pi_t \in \mathbb{F}_q^n$ that $\exists \pi_i : \delta(\pi_i, C) \geq \delta$, we have

$$\Pr_{\alpha_1,\ldots,\alpha_t \in \mathbb{F}_q}(\delta(\pi_0 + \alpha_1\pi_1 + \ldots + \alpha_t\pi_t, C) < \delta) \leq O\left(\frac{tn}{\varepsilon^2 q}\right).$$

  See Corollary 6 for the formal version.
- **Folded RS codes.** For $s, n, d \in \mathbb{N}^+, R = d/sn, s \geq L$, generator γ of $\mathbb{F}_q^\times$, and any appropriate sequence $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q^\times$. $C := \mathrm{FRS}_{n,d}^{(s,\gamma)}(\alpha_1, \ldots, \alpha_n)$ is the corresponding folded RS code over the alphabet $\mathbb{F}_q^s$ (see Section 4.2 for definition). Let $s > \lceil \frac{3}{\varepsilon^3} \rceil$ and $\delta < 1 - \sqrt{R + 2\varepsilon}$. For a batch of codewords $\pi_0, \ldots, \pi_t \in \mathbb{F}_q^n$ that $\exists \pi_i : \delta(\pi_i, C) \geq \delta$, we have

$$\Pr_{\alpha_1,\ldots,\alpha_t \in \mathbb{F}_q}(\delta(\pi_0 + \alpha_1\pi_1 + \ldots + \alpha_t\pi_t, C) < \delta) \leq O\left(\frac{tn}{\varepsilon^2 q}\right).$$

  See Corollary 7 and Corollary 8 for the formal versions.

Comparison to previous results. Previously, the best proximity gaps meeting the Johnson bound have been provided in [Ben+20b] for all RS codes, i.e., with arbitrary evaluation domains. More precisely, they provided that for an RS code C with block length n and rate R, if $\delta < 1 - \sqrt{R} - \eta$, then the bad-combining probability is bounded by

$$\frac{(Rn)^2}{\left(2\min(\eta, \frac{\sqrt{R}}{20})\right)^7 q} = O\left(\frac{n^2}{\eta^7 q}\right).$$

Results in [Ben+20b] hold for all RS codes, while our results (Corollary 6 and Corollary 7) only work for RS codes with specific evaluation domains. However, the above probability is quadratic in the code length, while our result is linear. For a fixed security level, our result implies that we can use a smaller field to decrease the cost of field operations.

[Ben+20b] provides the correlated agreement for RS codes. Our results support the stronger notion of *mutual correlated agreement*. The mutual correlated agreement for every linear code up to the 1.5 Johnson bound, i.e., $\delta < 1 - \sqrt[3]{R}$, has been shown in [GKL24][Zei24]. Our results for randomly punctured RS codes and folded RS codes meet the Johnson bound, i.e., $\delta < 1 - \sqrt{R}$. In application, improving this bound leads to fewer query times and higher protocol efficiency.

2 Preliminaries

Thoroughout the rest of this paper, we denote by $[n] := \{1, \ldots, n\}$ for all $n \in \mathbb{N}^+$. $\mathbb{F}_q$ is a finite field with q elements. Let $n \in \mathbb{N}^+$ and $\pi \in \mathbb{F}_q^n$ be a codeword of length n. For $j \in \{1, \ldots, n\}$, denote by $\pi[j]$ the element in π on the j-th position. Let $D \subseteq \{1, \ldots, n\}$ be a subdomain, denote by $\pi[D]$ the sub-vector of π containing elements on these positions.

2.1 Linear codes

Definition 1 (Linear code). *A linear code with message length k and codeword length n if it is a transformation from $\mathbb{F}_q^k$ to a linear subspace $C \subseteq \mathbb{F}_q^n$. The generator matrix $\mathbf{G} \in \mathbb{F}_q^{k \times n}$ of the linear code is a matrix whose rows form a basis of the code, i.e., for any codeword $c \in C$, there uniquely exists a message $\mathbf{m} \in \mathbb{F}_q^k$ such that $c = \mathbf{m} \cdot \mathbf{G}$.*

Definition 2 (Relative distance). *Let $\pi_1, \pi_2 \in \mathbb{F}_q^n$ be two codewords, where $n \in \mathbb{N}^+$. Define the relative Hamming distance between π_1 and π_2 to be*

$$\delta(\pi_1, \pi_2) := \frac{|\{j \in \{1, \ldots, n\} : \pi_1[j] \neq \pi_2[j]\}|}{n}.$$

Let $C \subseteq \mathbb{F}_q^n$ be a set of codewords with length n and $\pi \in \mathbb{F}_q^n$ be a codeword. Define the relative Hamming distance between π and C to be

$$\delta(\pi, C) := \min_{c \in C} \delta(\pi, c).$$

Definition 3 (Minimum relative distance). *Let $C \subseteq \mathbb{F}_q^n$ be a code with length n, where $n \in \mathbb{N}^+$. The minimum relative distance of C is defined as*

$$\Delta_C \triangleq \min_{c_1 \neq c_2 \in C} \delta(c_1, c_2).$$

Definition 4. *For a codeword $\pi \in \mathbb{F}_q^n$, let $\mathrm{Closest}(\pi, C) \in C$ denote the closest codeword to π (in terms of Hamming distance). If there are more than one codewords with the same minimal distance, choose the codeword with the smallest lexicographical order. That is, we fix a total order on the elements of $\mathbb{F}_q$ which induces a lexicographic order on $\mathbb{F}_q^n$.*

Definition 5 (List decoding). *Let $C \subseteq \mathbb{F}_q^n$ be a code. We say C is (p, L)-list-decodable if for every codeword $\pi \in \mathbb{F}_q^n$, we have*

$$|\{c \in C \mid \delta(c, \pi) \leq p\}| \leq L.$$

2.2 Correlated agreement

When considering the distance-preservation problem, we care about the overall performance of a series of codewords. Correlated agreement, introduced in [Ben+20b], is an important tool.

Let $C \subseteq \mathbb{F}_q^n$ be a linear code. Let $\pi_0, \ldots, \pi_t \in \mathbb{F}_q^n$ be a series of codewords. If $\pi_0, \ldots, \pi_t$ are not only close to C individually but also share a common large agreement domain, then we say $\pi_0, \ldots, \pi_t$ have a *correlated agreement.*

Definition 6 (Correlated agreement). *Let $\pi_0, \ldots, \pi_t \in \mathbb{F}_q^n$ be a sequence of codewords. Let $C \subseteq \mathbb{F}_q^n$ be a set of codewords. Let $0 < \delta \leq 1$. If there exists a subdomain $D \subseteq \{1, \ldots, n\}$ and $c_0, \ldots, c_t \in C$ satisfying*

- ***Density:*** *$|D| \geq (1 - \delta)n$, and*
- ***Agreement:*** *for all $i \in \{0, \ldots, t\}$, the codewords π_i and c_i agree on D.*

Then we say $\pi_0, \ldots, \pi_t$ have correlated agreement *with C of density $\geq 1 - \delta$.*

Definition 7 (Correlated distance). *Let $\pi_0, \ldots, \pi_t \in \mathbb{F}_q^n$ be a sequence of codewords. Let $C \subseteq \mathbb{F}_q^n$ be a set of codewords. Define the* correlated distance *of $\pi_0, \ldots, \pi_t$ with respect to C to be*

$$\delta_{\mathrm{corr}}(\{\pi_0, \ldots, \pi_t\}, C) := 1 - \max\left\{\frac{|D|}{n} : D \subseteq \{1, \ldots, n\}, \pi_0[D], \ldots, \pi_t[D] \in C[D]\right\}.$$

Definition 8. *Let $\pi_1, \pi_2 \in \mathbb{F}_q^n$ be two codewords. Function* Agree(π_1, π_2) *returns the positions where π_1 agrees with π_2, i.e.,*

$$\mathrm{Agree}(\pi_1, \pi_2) := \{j \in [n] \mid \pi_1[j] = \pi_2[j]\} \subseteq [n].$$

For a series of codewords and a given code, the following definition outlines all the correlated-agree domains that are large enough between them.

Definition 9 (Maximal δ-correlated-agree domains). *Let $0 \leq \delta \leq 1$. Let $C \subseteq \mathbb{F}_q^n$ be a linear code. Let $\pi_1, \ldots, \pi_l \in \mathbb{F}_q^n$ be a series of codewords with $l \geq 1$. Let $D \subseteq \{1, \ldots, n\}$ be a domain satisfying:*

- ***Density:*** *$|D| \geq (1 - \delta)n$;*
- ***Correlated agreement:*** *For $i \in \{1, \ldots, l\}$, $\exists c_i \in C$, such that $\pi_i[D] = c_i[D]$;*
- ***Maximality:*** *If $D \subsetneq D'$, then $\exists i \in \{1, \ldots, l\}$, for all $c \in C$, $\pi_i[D'] \neq c[D']$.*

We define such a domain D as a maximal δ-correlated-agree domain between $\{\pi_1, \ldots, \pi_l\}$ and C. Denote the set of all of the maximal δ-correlated-agree domains between $\{\pi_1, \ldots, \pi_l\}$ and C as

$$\mathcal{A}_{\delta, \{\pi_1, \ldots, \pi_l\}, C} \triangleq \{D_1, \ldots, D_m\}.$$

Notice that $\mathcal{A}_{\delta, \{\pi_1, \ldots, \pi_l\}, C}$ is uniquely determined, and when $\delta_{\mathrm{corr}}(\{\pi_1, \ldots, \pi_l\}, C) > \delta$, $\mathcal{A}_{\delta, \{\pi_1, \ldots, \pi_l\}, C}$ is empty.

3 From list-decodability to proximity gaps

In this section, we prove our main result: a general method for establishing the mutual correlated agreement of a linear code with good list-decodability. In Section 3.1, we establish the proximity gaps for combining two codewords. In Section 3.2, we generalize this result to a batch of codewords and offer two versions of the combining structure: the plain combining and the tensor combining.

We establish the proximity gap for a linear code C by upper bounding the number of (plain) bad-combining points as defined below:

Definition 10 (Bad-combining points). *Let $\mathbb{F}_q$ be a finite field. Let $C \subseteq \mathbb{F}_q^n$ be a linear code. Denote by Δ_C the minimum relative distance of C. Let $\delta \in (0, 1-\Delta_C)$. Let $\pi_0, \pi_1, \ldots, \pi_t \in \mathbb{F}_q^n$ be codewords with maximal δ-correlated-agree domains $\mathcal{A}_{\delta,\{\pi_0,\ldots,\pi_t\},C} = \{D_1, \ldots, D_m\}$, where $t \geq 1$. Define the* bad combining points *of $(\pi_0, \ldots, \pi_t)$ (with respect to the linear code C) to be*

$$\mathrm{Bad}_C(\pi_0, \ldots, \pi_t, \delta) :=$$
$$\left\{ \langle \alpha_1, \ldots, \alpha_t \rangle \in \mathbb{F}_q^t : \begin{array}{l} \exists c \in C, \ \textit{such that} \\ \delta(\pi_0 + \alpha_1\pi_1 + \ldots + \alpha_t\pi_t, c) \leq \delta \ \textit{and} \\ \mathrm{Agree}(\pi_0 + \alpha_1\pi_1 + \ldots + \alpha_t\pi_t, c) \notin \mathcal{A}_{\delta,\{\pi_0,\ldots,\pi_t\},C}. \end{array} \right\}$$

We omit C and use $\mathrm{Bad}(\pi_0, \ldots, \pi_t, \delta)$ when the corresponding linear code is clear.

3.1 Proximity gaps for combining two codewords

In this subsection, we prove the following theorem, which is the first main result of the paper.

Theorem 3. *Let $C \subseteq \mathbb{F}_q^n$ be a linear code with minimum relative distance Δ_C. $p < \Delta_C$ and suppose C is (p, L) list-decodable. $\varepsilon > 0, \delta \leq 1 - \sqrt{1 - p + \varepsilon}$. Let $\pi_1, \pi_2 \in \mathbb{F}_q^n$ two codewords and $\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C} = \{D_1, \ldots, D_m\}$ be all the maximal δ-correlated agree domains between $\{\pi_1, \pi_2\}$ and C. Then we have*

$$|\mathrm{Bad}(\pi_1, \pi_2, \delta)| < L^2pn + \frac{1}{\varepsilon}.$$

Remark 1. Theorem 3 implies the *mutual correlated agreement* (as defined in Definition 4.9 in [Arn+25]) of C. This property is a strengthening of correlated agreement and plays a key role in establishing the soundness of certain protocols, including those in [Arn+25] and [Bün+25].

Partition $\mathrm{Bad}(\pi_1, \pi_2, \delta)$. To bound the number of elements in $\mathrm{Bad}(\pi_1, \pi_2, \delta)$, we need to partition $\mathrm{Bad}(\pi_1, \pi_2, \delta)$ into two parts and bound the number of elements in each part respectively.

Informally, we partition the bad-combining points into two parts: (1) the combining result is improved from some *low-error* codewords, or (2) the combining result is gained by some *high-error* codewords. More precisely, for an

$\alpha \in \mathrm{Bad}(\pi_1, \pi_2, \delta)$, if $\exists c$ such that $\delta(\pi_1 + \alpha\pi_2, c) \le \delta$, then we have either $\exists D \in \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$ such that $\mathrm{Agree}(\pi_1 + \alpha\pi_2, c) \supsetneq D$ or $\forall D \in \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$, $|\mathrm{Agree}(\pi_1 + \alpha\pi_2, c) \cap D| \le (1 - \Delta_C)n$. Define:

$$\mathrm{Bad}(\pi_1, \pi_2, \delta)^1 := \left\{ \alpha \in \mathrm{Bad}(\pi_1, \pi_2, \delta) : \begin{array}{l} \exists c \in C, D \in \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C} \text{ such that} \\ \delta(\pi_1 + \alpha\pi_2, c) \le \delta \text{ and} \\ D \subsetneq \mathrm{Agree}(\pi_1 + \alpha\pi_2, c) \end{array} \right\}$$

and

$$\mathrm{Bad}(\pi_1, \pi_2, \delta)^2 = \mathrm{Bad}(\pi_1, \pi_2, \delta) \setminus \mathrm{Bad}(\pi_1, \pi_2, \delta)^1.$$

Bound the first part. To bound $|\mathrm{Bad}(\pi_1, \pi_2, \delta)^1|$, we further define some subsets of $\mathrm{Bad}(\pi_1, \pi_2, \delta)^1$. Denote by $\mathcal{D} := \{D_1, \ldots, D_l\}$ all the maximal p-correlated-agree domains between $\{\pi_1, \pi_2\}$ and C. Since $1 - p \le \sqrt{1 - p + \varepsilon} \le 1 - \delta$, $\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C} \subseteq \mathcal{D}$. For convenience, let the first m elements in $\mathcal{D}$ be the maximal δ-correlated agree domains.

We show that the number of subdomains in $\mathcal{D}$ is bounded by L^2, where L is the list size. Consequently, we obtain an upper bound on the number of subdomains in $\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$.

Claim. $l \le L^2$.

Proof. Let

$$\{f_1, \ldots, f_{l_1}\} \subseteq C$$

be all the codewords that p-close to π_1, i.e., we have $\delta(\pi_1, f_i) \le p$ for every $i \in [l_1]$ and $\delta(\pi_1, f) > p$ for any $f \in C \setminus \{f_1, \ldots, f_{l_1}\}$. Let $\{B_1, \ldots, B_{l_1}\}$ be the corresponding agree domains, i.e.,

$$B_i = \mathrm{Agree}(\pi_1, f_i), \text{ for } i = 1, \ldots, l_1.$$

Similarly, we define $\{g_1, \ldots, g_{l_2}\}$ and $\{C_1, \ldots, C_{l_2}\}$ to be the list-decoding result and agree domains of π_2 respectively. Since C is (p, L) list-decodable, we have $l_1 \le L$ and $l_2 \le L$.

$\forall D \in \mathcal{D}$, $\exists c \in C$ such that $\pi_1[D] = c[D]$. Since $|D| \ge (1 - p)n$, $\exists B_i \in \{B_1, \ldots, B_{l_1}\}$ such that $D \subseteq B_i$. Similarly, $\exists C_j \in \{C_1, \ldots, C_{l_2}\}$ such that $D \subseteq C_j$. Thus, $D \subseteq B_i \cap C_j$. On the other hand, $B_i \cap C_j$ is a correlated agree domain between $\{\pi_1, \pi_2\}$ and C. Since D is maximal, $B_i \cap C_j \subseteq D$. Thus, $D = B_i \cap C_j$.

We have proved all the sets in $\mathcal{D}$ are of the form $B_i \cap C_j$, where $B_i \in \{B_1, \ldots, B_{l_1}\}, C_j \in \{C_1, \ldots, C_{l_2}\}$. Notice that there are fewer than $l_1 l_2 \le L^2$ different intersections over the choice of agree domains. As a result, $|\mathcal{D}| \le L^2$.

For $D \in \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$, let

$$\mathrm{Bad}(\pi_1, \pi_2, \delta)^1_D := \left\{ \alpha \in \mathrm{Bad}(\pi_1, \pi_2, \delta)^1 : \begin{array}{l} \exists c \in C \text{ such that} \\ \delta(\pi_1 + \alpha\pi_2, c) \le \delta \text{ and} \\ D \subsetneq \mathrm{Agree}(\pi_1 + \alpha\pi_2, c) \end{array} \right\}.$$

We have

$$
\begin{aligned}
|\mathrm{Bad}(\pi_1,\pi_2,\delta)^1| &= |\bigcup_{D\in\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}} \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D| \\
&\le \sum_{D\in\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}} |\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D| \\
&\le L^2 \cdot \max_{D\in\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}} |\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D|.
\end{aligned}
$$

If we can bound the number of elements in $\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$ for any $D \in \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$, we finish the proof of the first part. The proof is based on the following observation in [BKS18][Ben+20a].

Lemma 1. *For any distinct $\alpha_1,\alpha_2 \in \mathbb{F}_q$, let $B_{\alpha_1} \subseteq [n]$ be a maximal agree-domain between $\pi_1+\alpha_1\pi_2$ and C and $B_{\alpha_2} \subseteq [n]$ be a maximal agree-domain between $\pi_1+\alpha_2\pi_2$ and C. Then $B_{\alpha_1}\cap B_{\alpha_2}$ is a correlated-agree-domain between $\{\pi_1,\pi_2\}$ and C, i.e., there exist codewords $c_1,c_2 \in C$ such that*

$$
\pi_1[B_{\alpha_1}\cap B_{\alpha_2}] = c_1[B_{\alpha_1}\cap B_{\alpha_2}] \text{ and } \pi_2[B_{\alpha_1}\cap B_{\alpha_2}] = c_2[B_{\alpha_1}\cap B_{\alpha_2}]. \quad (3)
$$

Moreover, if $|B_{\alpha_1}\cap B_{\alpha_2}| > (1-\Delta_C)n$, codewords c_1 and c_2 are uniquely determined and we have

$$
B_{\alpha_1} = \mathrm{Agree}(\pi_1+\alpha_1\pi_2, c_1+\alpha_1 c_2) \text{ and } B_{\alpha_2} = \mathrm{Agree}(\pi_1+\alpha_2\pi_2, c_1+\alpha_2 c_2).
$$

This implies the following property of the intersection of agree domains [GKL24]:

Corollary 1. *Let $\alpha,\beta \in \mathbb{F}_q$ be distinct combining points. Let $B_\alpha, B_\beta \subseteq [n]$ be maximal agree-domains such that $|B_\alpha \cap B_\beta| > (1-\Delta_C)n$. Then $B_\alpha\cap B_\beta$ is a maximal correlated-agree-domain between $\{\pi_1,\pi_2\}$ and C.*

The following lemma states that for any two distinct elements in $\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$, the corresponding intersection is exactly D.

Lemma 2. *For any fixed $D \in \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$, let $\alpha,\beta \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$ be two distinct elements. $c_\alpha, c_\beta \in C$ are the codewords such that $\mathrm{Agree}(\pi_1+\alpha\pi_2, c_\alpha) \supsetneq D$ and $\mathrm{Agree}(\pi_1+\beta\pi_2, c_\beta) \supsetneq D$. Then we have $\mathrm{Agree}(\pi_1+\alpha\pi_2, c_\alpha)\cap\mathrm{Agree}(\pi_1+\beta\pi_2, c_\beta) = D$.*

Proof. For convenience, denote by $A_\alpha = \mathrm{Agree}(\pi_1+\alpha\pi_2, c_\alpha)$ and $A_\beta = \mathrm{Agree}(\pi_1+\beta\pi_2, c_\beta)$ the agree-domains. According to the definition of c_α, c_β, we have $D \subseteq A_\alpha \cap A_\beta$.

On the other hand, Lemma 1 tells us there exist $c_1, c_2 \in C$ that

$$
\pi_1[A_\alpha\cap A_\beta] = c_1[A_\alpha\cap A_\beta] \text{ and } \pi_2[A_\alpha\cap A_\beta] = c_2[A_\alpha\cap A_\beta].
$$

Since $D \subseteq A_\alpha\cap A_\beta$, $|A_\alpha\cap A_\beta| \ge |D| \ge (1-p)n > (1-\Delta_C)n$, c_1, c_2 are uniquely determined. Thus, $A_\alpha\cap A_\beta$ is a p-correlated agree domain between $\{\pi_1,\pi_2\}$ and C. Then we have $A_\alpha\cap A_\beta \subseteq D$ because D is maximal.

According to the definition of $\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$, A_α is a strict superset of D. By counting the elements in $A_\alpha \setminus D$, we can bound the number of elements in $\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$.

Corollary 2. *$|\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D| \le pn$.*

Proof. For any two distinct $\alpha,\beta \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$, Lemma 2 implies

$$(A_\alpha \setminus D) \cap (A_\beta \setminus D) = \emptyset. \tag{4}$$

For every $\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$, we have $A_\alpha \setminus D \neq \emptyset$ according to the definition of $\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$.

Then the following inequality holds:

$$\begin{aligned}
|[n]\setminus D| &\ge \left| \bigcup_{\alpha\in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D} A_\alpha \setminus D \right| \\
&= \sum_{\alpha\in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D} |A_\alpha \setminus D| && \text{by (4)} \\
&\ge \sum_{\alpha\in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D} 1 = |\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D|
\end{aligned}$$

For the left side of the inequality, we have $|[n]\setminus D| = n-|D| < n-(1-p)n = pn$. We have an upper bound of $\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D$ that $|\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D| \le pn$.

Bound the second part. For any two distinct $\alpha,\beta \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2$, the corresponding A_α, A_β satisfy

1. $|A_\alpha| \ge (1-\delta)n, |A_\beta| \ge (1-\delta)n$ according to the definition of $\mathrm{Bad}(\pi_1,\pi_2,\delta)$;
2. $|A_\alpha \cap A_\beta| \le (1-p)n$ according to Lemma 1 and the definition of $\mathrm{Bad}(\pi_1,\pi_2,\delta)^2$.

By counting over the coordinate positions, we can derive an upper bound on the size of the set $\mathrm{Bad}(\pi_1,\pi_2,\delta)^2$.

Lemma 3. *$|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| < \frac{1}{\varepsilon}$.*

Proof. For any $\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2 \subseteq \mathrm{Bad}(\pi_1,\pi_2,\delta)$, $\exists c \in C$ such that $|\mathrm{Agree}(\pi_1+\alpha\pi_2, c)| \ge (1-\delta)n \ge \sqrt{1-p+\varepsilon}\cdot n$ and $\mathrm{Agree}(\pi_1+\alpha\pi_2,c) \notin \mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}$ according to the definition of $\mathrm{Bad}(\pi_1,\pi_2,\delta)$. For convenience, denoted by $A_\alpha = \mathrm{Agree}(\pi_1+\alpha\pi_2,c)$ the agree domain.

For any distinct $\alpha,\beta \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2$, we claim that $|A_\alpha \cap A_\beta| < (1-p)n$. This is because $A_\alpha \cap A_\beta$ is a correlated agree domain between $\{\pi_1,\pi_2\}$ and C according to Lemma 1. If $|A_\alpha \cap A_\beta| \ge (1-p)n > (1-\Delta_C)n$, then $\exists D \in \mathcal{D}$ such that $A_\alpha \cap A_\beta \subseteq D$ according to the definition of $\mathcal{D}$. By Corollary 1, $A_\alpha \cap A_\beta$ is a maximal correlated agree domain, thus, $A_\alpha \cap A_\beta = D$. Since A_α is not a correlated agree domain between $\{\pi_1,\pi_2\}$ and C, we have $A_\alpha \neq D$. Thus, $A_\alpha \supsetneq D$, and $\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^1$, a contradiction.

We count the elements in $\bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha$. For any $x \in \bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha$, denote by $d(x)$ the count of x, i.e., the number of A_α containing x. We have

$$\begin{aligned}
\sum_{x \in A_\alpha} d(x) &= \sum_{\beta \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} |A_\alpha \cap A_\beta| \\
&= |A_\alpha| + \sum_{\beta \neq \alpha} |A_\alpha \cap A_\beta| \\
&< |A_\alpha| + (|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| - 1)(1-p)n. \qquad (5)
\end{aligned}$$

Summing over all sets A_α, we have

$$\begin{aligned}
\sum_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} \sum_{x \in A_\alpha} d(x) &= \sum_{x \in \bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha} d(x)^2 \\
&\geq \frac{1}{\left|\bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha\right|} \left(\sum_{x \in \bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha} d(x) \right)^2 \\
&= \frac{1}{\left|\bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha\right|} \left(\sum_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} |A_\alpha| \right)^2, \qquad (6)
\end{aligned}$$

where the first inequality is by Cauchy–Schwarz inequality. Let $r = \left|\bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha\right|$, $s = \sum_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} |A_\alpha| \geq |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \sqrt{1-p+\varepsilon} \cdot n$. Combining (5) and (6), we have

$$\begin{aligned}
& s + |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2|(|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| - 1)(1-p)n > \frac{s^2}{r} \\
\Rightarrow\ & r > \frac{s^2}{s + |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \left(|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| - 1\right)(1-p)n}.
\end{aligned}$$

Notice that the right side of the above inequality is monotonically increasing with respect to s when $s \geq 0$. Thus, we have

$$r > \frac{(1-p+\varepsilon)|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| n}{\sqrt{1-p+\varepsilon} + \left(|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| - 1\right)(1-p)}.$$

Since $\bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha$ is a subset of $[n]$, we have

$$\begin{aligned}
& n \geq |\bigcup_{\alpha \in \mathrm{Bad}(\pi_1,\pi_2,\delta)^2} A_\alpha| > \frac{(1-p+\varepsilon)|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \cdot n}{\sqrt{1-p+\varepsilon} + \left(|\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| - 1\right)(1-p)} \\
\Rightarrow & |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| < \frac{\sqrt{1-p+\varepsilon} - (1-p)}{\varepsilon} < \frac{1}{\varepsilon}.
\end{aligned}$$

Now, we can prove Theorem 3 by Claim 3.1, Corollary 2, and Lemma 3:

Proof (Proof of Theorem 3).

$$\begin{aligned}|\mathrm{Bad}(\pi_1,\pi_2,\delta)| &= |\mathrm{Bad}(\pi_1,\pi_2,\delta)^1 \cup \mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \\ &= |\mathrm{Bad}(\pi_1,\pi_2,\delta)^1| + |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \\ &= |\bigcup_{D\in\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}} \mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D| + |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \\ &\le L^2 \cdot \max_{D\in\mathcal{A}_{\delta,\{\pi_1,\pi_2\},C}} |\mathrm{Bad}(\pi_1,\pi_2,\delta)^1_D| + |\mathrm{Bad}(\pi_1,\pi_2,\delta)^2| \\ &< L^2pn + \frac{1}{\varepsilon}.\end{aligned}$$

3.2 Proximity gaps for combining a batch of codewords

The main theorem can be extended to combine a batch of codewords. The proof uses induction and follows [GKL24]. To make it easy to read, we provide the detailed proof in the Appendix A.

Corollary 3. *Let $C \subseteq \mathbb{F}_q^n$ be a linear code with minimum relative distance Δ_C. $p < \Delta_C$ and suppose C is (p, L) list-decodable. $\delta \le 1 - \sqrt{1-p+\varepsilon}$. $\pi_0, \ldots, \pi_t \in \mathbb{F}^n$ are $t+1$ codewords with maximal δ-correlated agree domains $\mathcal{A}_{\delta,\{\pi_0,\ldots,\pi_t\},C} = \{D_1, \ldots, D_m\}$. Then we have*

$$\Pr_{\langle\alpha_1,\ldots,\alpha_t\rangle\in\mathbb{F}_q}\left(\langle\alpha_1,\ldots,\alpha_t\rangle \in \mathrm{Bad}(\pi_0,\ldots,\pi_t,\delta)\right) < \frac{t}{q}\cdot\left(L^2pn + \frac{1}{\varepsilon}\right).$$

We further provide the result for codewords combining in a tensor structure. This type of structure arises in certain proximity testing protocols that employ *folding* as a core technique, notably in FRI[Ben+18]. More precisely, performing the 2-wise folding operation t times on a codeword with initial code-length $2^t n$ (Reed-Solomon codes over a smooth evaluation domain) with folding parameters $\beta_1, \ldots, \beta_t$ can be interpreted as a tensor combination of 2^t codewords with code-length n.

Let $C \subseteq \mathbb{F}_q^n$ be a set of codewords. For a series of codewords $\pi_1, \ldots, \pi_{2^t} \in \mathbb{F}_q^n, t \in \mathbb{N}$, let $\mathcal{A}_{\delta,\{\pi_1,\ldots,\pi_{2^t}\},C} = \{D_1, \ldots, D_m\}$ be the maximal δ-correlated agree domains between them and C. Define the *tensor combining* of them to be:

$$\langle\pi_1,\ldots,\pi_{2^t}\rangle \cdot \otimes_{i=1}^t \langle 1, \beta_i\rangle,$$

where $\otimes_{i=1}^t\langle 1,\beta_i\rangle = \langle 1,\beta_1\rangle \otimes \langle 1,\beta_2\rangle \otimes \ldots \otimes \langle 1,\beta_t\rangle$ denotes the tensor product and $\cdot$ denotes the inner product of vectors.

Define the set of bad folding points to be

$$\mathrm{Bad}_{\mathrm{T}}(\pi_1,\ldots,\pi_{2^t},\delta) :=$$
$$\left\{\langle\beta_1,\ldots,\beta_t\rangle \in \mathbb{F}_q^t : \begin{array}{l}\exists c \in C, \text{ such that} \\ \delta\left(\langle\pi_1,\ldots,\pi_{2^t}\rangle \cdot \otimes_{i=1}^t\langle 1,\beta_i\rangle, c\right) \le \delta \text{ and} \\ \mathrm{Agree}(\langle\pi_1,\ldots,\pi_{2^t}\rangle \cdot \otimes_{i=1}^t\langle 1,\beta_i\rangle, c) \notin \mathcal{A}_{\delta,\{\pi_1,\ldots,\pi_{2^t}\},C}\end{array}\right\}.$$

Notice that when $t = 1$, the above definition equals the common bad folding points.

Corollary 4. *Let $C \subseteq \mathbb{F}_q^n$ be a linear code with minimum relative distance Δ_C. $p < \Delta_C$ and suppose C is (p, L) list-decodable. $\delta \le 1 - \sqrt{1 - p + \varepsilon}$. $\pi_1, \dots, \pi_{2^t} \in \mathbb{F}^n$ are 2^t codewords with maximal δ-correlated agree domains $\mathcal{A}_{\delta,\{\pi_1,\dots,\pi_{2^t}\},C} = \{D_1, \dots, D_m\}$. Then we have*

$$\Pr_{\langle \beta_1,\dots,\beta_t\rangle \in \mathbb{F}_q^t}(\langle \beta_1, \dots, \beta_t\rangle \in \mathrm{Bad}_{\mathrm{T}}(\pi_1, \dots, \pi_{2^t}, \delta)) < \frac{2^t - 1}{q} \cdot \left(L^2 pn + \frac{1}{\varepsilon}\right).$$

Proof. We prove this by induction. When $t = 1$, the result holds according to Theorem 3. Suppose this holds for less than 2^{t-1} codewords, we prove the result for 2^t codewords. Notice that

$$\langle \pi_1, \dots, \pi_{2^t}\rangle \cdot \otimes_{i=1}^{t} \langle 1, \beta_i\rangle = \langle \pi_1 + \beta_t \pi_2, \pi_3 + \beta_t \pi_4, \dots, \pi_{2^t-1} + \beta_t \pi_{2^t}\rangle \cdot \otimes_{i=1}^{t-1} \langle 1, \beta_i\rangle,$$

we partition the codewords into 2^{t-1} groups:

$$\{\pi_{2i-1}, \pi_{2i}\} \text{ for } 1 \le i \le 2^{t-1}.$$

Then $\Pr_{\beta_t \in \mathbb{F}_q}(\beta_t \in \mathrm{Bad}(\pi_{2i-1}, \pi_{2i}, \delta)) < \frac{1}{q} \cdot \left(L^2 pn + \frac{1}{\varepsilon}\right)$ for every $i \in [2^{t-1}]$. Denote by A_i the case that $\beta_t \in \mathrm{Bad}(\pi_{2i-1}, \pi_{2i}, \delta)$, we have

$$\Pr\left(\bigvee_{i=1}^{2^{t-1}} A_i\right) \le \sum_{i=1}^{2^{t-1}} \Pr(A_i) < \frac{2^{t-1}}{q} \cdot \left(L^2 pn + \frac{1}{\varepsilon}\right).$$

Suppose A_i does not happen for all $i \in [2^{t-1}]$, we claim that

$$\mathcal{A}_{\delta,\{\pi_1+\beta_t\pi_2,\dots,\pi_{2^t-1}+\beta_t\pi_{2^t}\},C} = \{D_1, \dots, D_m\}.$$

$\{D_1, \dots, D_m\} \subseteq \mathcal{A}_{\delta,\{\pi_1+\beta_t\pi_2,\dots,\pi_{2^t-1}+\beta_t\pi_{2^t}\},C}$ is obvious. We prove the other direction. Otherwise, suppose $\exists D \in \mathcal{A}_{\delta,\{\pi_1+\beta_t\pi_2,\dots,\pi_{2^t-1}+\beta_t\pi_{2^t}\},C}$ such that $D \notin \{D_1, \dots, D_m\}$. Then $\pi_1 + \beta_t \pi_2, \dots, \pi_{2^t-1} + \beta_t \pi_{2^t}$ agree with C on D. For every $i \in [2^{t-1}]$, D is a correlated agree domain between $\{\pi_{2i-1}, \pi_{2i}\}$ and C since A_i does not happen. Thus, D is a correlated agree domain between $\{\pi_1, \dots, \pi_{2^t}\}$ and C, a contradiction.

For a fixed β_t that A_i does not happen for every $i \in [2^{t-1}]$, we have

$$\begin{aligned} &\langle \beta_1, \dots, \beta_t\rangle \in \mathrm{Bad}_{\mathrm{T}}(\pi_1, \dots, \pi_{2^t}, \delta) \\ \Rightarrow &\langle \beta_1, \dots, \beta_{t-1}\rangle \in \mathrm{Bad}_{\mathrm{T}}(\pi_1 + \beta_t \pi_2, \dots, \pi_{2^t-1} + \beta_t \pi_{2^t}, \delta). \end{aligned}$$

This is because for a fixed β_t that A_i does not happen for every $i \in [2^{t-1}]$ and $\langle \beta_1, \dots, \beta_{t-1}\rangle \notin \mathrm{Bad}_{\mathrm{T}}(\pi_1 + \beta_t \pi_2, \dots, \pi_{2^t-1} + \beta_t \pi_{2^t}, \delta)$ imply $\langle \beta_1, \dots, \beta_t\rangle \notin \mathrm{Bad}_{\mathrm{T}}(\pi_1, \dots, \pi_{2^t}, \delta)$.

Thus,

$$
\begin{aligned}
&\Pr_{\beta_1,\ldots,\beta_t\in\mathbb{F}_q}\left(\langle\beta_1,\ldots,\beta_t\rangle\in\mathrm{Bad_T}(\pi_1,\ldots,\pi_{2^t},\delta)\right)\\
\leq&\Pr_{\beta_1,\ldots,\beta_t\in\mathbb{F}_q}\left(\langle\beta_1,\ldots,\beta_t\rangle\in\mathrm{Bad_T}(\pi_1,\ldots,\pi_{2^t},\delta)\mid\bigwedge_{i=1}^{2^{t-1}}\overline{A}_i\right)+\Pr_{\beta_t\in\mathbb{F}_q}\big(\bigvee_{i=1}^{2^{t-1}}A_i\big)\\
\leq&\Pr_{\beta_1,\ldots,\beta_t\in\mathbb{F}_q}\left(\langle\beta_1,\ldots,\beta_{t-1}\rangle\in\mathrm{Bad_T}(\pi_1+\beta_t\pi_2,\ldots,\pi_{2^t-1}+\beta_t\pi_{2^t},\delta)\mid\bigwedge_{i=1}^{2^{t-1}}\overline{A}_i\right)\\
&+\Pr_{\beta_t\in\mathbb{F}_q}\big(\bigvee_{i=1}^{2^{t-1}}A_i\big)\\
<&\frac{2^{t-1}-1}{q}\cdot\left(L^2pn+\frac{1}{\varepsilon}\right)+\frac{2^{t-1}}{q}\cdot\left(L^2pn+\frac{1}{\varepsilon}\right)\\
=&\frac{2^t-1}{q}\cdot\left(L^2pn+\frac{1}{\varepsilon}\right).
\end{aligned}
$$

4 Proximity gaps for RS codes

This section applies our main result to two specific Reed-Solomon codes with list-decodability up to capacity. More precisely, we provide linear proximity gaps for randomly punctured RS codes and folded RS codes under the Johnson bound.

4.1 Proximity gaps for randomly punctured RS codes

The following theorem in [AGL24] proves that with high possibility (over the choice of the evaluation domain), a Reed-Solomon code with rate R is average-radius list-decodable when $p < 1-R$. Average-radius list-decodability is stronger than list-decodability.

Theorem 4 ([AGL24]). *Let $\varepsilon\in(0,1), L\geq 2$ and q be a prime power such that $q\geq n+k\cdot 2^{10L/\varepsilon}$. Then with probability at least $1-2^{-Ln}$, a randomly punctured Reed-Solomon code of block length and rate $R=k/n$ over $\mathbb{F}_q$ is $(\frac{L}{L+1}(1-R-\varepsilon), L)$ average-radius list-decodable.*

Remark 2. The list-decodability of random linear codes has also been established in [AGL24]. By applying our framework to this setting, we obtain proximity gaps for random linear codes as well.

Taking $L=O(1/\varepsilon)$ in Theorem 4, [AGL24] also proves that RS codes achieve list-decoding capacity over linear-sized alphabets.

Corollary 5 ([AGL24]). *Let $\varepsilon\in(0,1), L\geq 2$ and q be a prime power such that $q\geq n+k\cdot 2^{O(1/\varepsilon^2)}$. Then with probability at least $1-2^{-\Omega(n/\varepsilon)}$, a randomly punctured Reed-Solomon code of block length and rate $R=k/n$ over $\mathbb{F}_q$ is $(1-R-\varepsilon, O(\frac{1}{\varepsilon}))$ average-radius list-decodable.*

We provide the following proximity gaps (mutually correlated agreement) for randomly punctured RS codes by directly applying our main result to Corollary 5.

Corollary 6. *Let $\varepsilon \in (0,1)$ and q be a prime power such that $q \geq n+k\cdot 2^{O(1/\varepsilon^2)}$. Then with probability at least $1-2^{-\Omega(n/\varepsilon)}$ over the choice of evaluation points of C with block length n and rate $R = k/n$ over $\mathbb{F}_q$ is $(1-R-\varepsilon, O(\frac{1}{\varepsilon}))$ average-radius list-decodable.*

At this time, for any given series of codewords $\pi_0, \ldots, \pi_t$ and $\delta \leq 1-\sqrt{R+2\varepsilon}$, we have

$$\Pr_{\langle\beta_1,\ldots,\beta_t\rangle\in\mathbb{F}_q^t}(\langle\beta_1,\ldots,\beta_t\rangle \in \mathrm{Bad}_C(\pi_0,\ldots,\pi_t,\delta)) \leq O\left(\frac{n}{\varepsilon^2 q}\right).$$

4.2 Proximity gaps for folded RS codes

We also apply our main result to folded Reed-Solomon codes, initially introduced by Krachkovsky in [Kra03]. They are RS codes with structured evaluation domains. Formally, let $s, n > 0$ be two parameters, $d > 0$ be the degree bound, and $\mathbb{F}_q$ be a finite field with $q > ns > d$. $\gamma \in \mathbb{F}_q^\times$ is a generator of $\mathbb{F}_q^\times$. Given a series of $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$, the corresponding (s,γ)-folded RS over the alphabet $\mathbb{F}_q^s$ of block length n and rate $R = d/ns$ can be defined as

$$\mathrm{FRS}_{n,d}^{(s,\gamma)}(\alpha_1,\ldots,\alpha_n) := \{\mathcal{C}(f) \mid f \in \mathbb{F}_q^{<d}[X]\} \subseteq (\mathbb{F}_q^s)^n,$$

where $\mathcal{C}(f) := (F_1,\ldots,F_n)$ with $F_i := \big(f(\alpha_i), f(\gamma\alpha_i), \ldots, f(\gamma^{s-1}\alpha_i)\big)$ denotes the encoder of this code.

Recently, an exciting work by Chen and Zhang [CZ24] has shown that any *appropriate* folded RS code of block length n and rate R over the alphabet $\mathbb{F}_q^s$ of length n is list-decodable up to capacity, here appropriate refers to $\alpha_i\gamma^j, i \in [n], j \in \{0\}\cup[s-1]$ are distinct.

Theorem 5 ([CZ24]). *For any $L \geq 1, s, n, d \in \mathbb{N}^+, R = \frac{d}{sn}, q > n, s \geq L$, generator γ of $\mathbb{F}_q^\times$, and any appropriate sequence $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q^\times$, the corresponding folded RS code $\mathrm{FRS}_{n,d}^{(s,\gamma)}(\alpha_1,\ldots,\alpha_n)$ is $\left(\frac{L}{L+1}\left(1-\frac{sR}{s-L+1}\right), L\right)$ list-decodable.*

This is the first explicit construction of $(1-R-\varepsilon, O(1/\varepsilon))$ (average radius) list-decodable codes of rate R with polynomial-sized alphabets. Applying Corollary 3 to folded RS codes, we have the following proximity gaps for folded RS codes:

Corollary 7 (Proximity gaps for folded RS codes). *For any $L \geq 1, s, n, d \in \mathbb{N}^+, R = d/sn, q > n, s \geq L$, generator γ of $\mathbb{F}_q^\times$, and any appropriate sequence $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q^\times$. $C := \mathrm{FRS}_{n,d}^{(s,\gamma)}(\alpha_1,\ldots,\alpha_n)$ is the corresponding folded RS code over the alphabet $\mathbb{F}_q^s$. $\varepsilon > 0$ and $\delta \leq 1-\sqrt{1-\frac{L}{L+1}\left(1-\frac{sR}{s-L+1}\right)+\varepsilon}$.*

$\pi_0, \ldots, \pi_t \in (\mathbb{F}_q^s)^n$ are $t+1$ codewords with maximal δ-correlated agree domains $\mathcal{A}_{\delta,\{\pi_0,\ldots,\pi_t\},C} = \{D_1, \ldots, D_m\}$. Then we have

$$\begin{aligned} &\Pr_{\langle \beta_1,\ldots,\beta_t\rangle \in \mathbb{F}_q^t}(\langle \beta_1, \ldots, \beta_t\rangle \in \mathrm{Bad}\,(\pi_0, \ldots, \pi_t, \delta)) \\ &< \frac{t}{q} \cdot \left(\frac{L^3 n}{L+1} \left(1 - \frac{sR}{s-L+1}\right) + \frac{1}{\varepsilon}\right). \end{aligned}$$

We denote the above upper bound of the probability by

$$\mathbf{err}_{\mathrm{comb}}(t, R, s, n, L, \varepsilon, q) := \frac{t}{q} \cdot \left(\frac{L^3 n}{L+1} \left(1 - \frac{sR}{s-L+1}\right) + \frac{1}{\varepsilon}\right).$$

Following the approach of Corollary 2.21 in [CZ24], we provide some results with differently chosen s.

Corollary 8. *For any $L \geq 1, s, n, d \in \mathbb{N}^+, R = d/sn, q > n, s \geq L$, generator γ of $\mathbb{F}_q^\times$, and any appropriate sequence $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q^\times$. $C := \mathrm{FRS}_{n,d}^{(s,\gamma)}(\alpha_1, \ldots, \alpha_n)$ is the corresponding folded RS code over the alphabet $\mathbb{F}_q^s$. $\pi_0, \ldots, \pi_t \in (\mathbb{F}_q^s)^n$ are $t+1$ codewords. Then we have*

1. *When $L = \lfloor \frac{1-R}{\varepsilon} \rfloor, s > \frac{L(L-1)R}{\varepsilon L - (1-R-\varepsilon)} + L - 1$, and $\delta < 1 - \sqrt{R + 2\varepsilon}$, we have*

$$\mathbf{err}_{\mathrm{comb}}(t, R, s, n, L, \varepsilon, q) < \frac{t}{q} \cdot \left(\frac{(1-R)^3}{\varepsilon^2} n + \frac{1}{\varepsilon}\right). \tag{7}$$

2. *When $L = \lceil \frac{1-R}{\varepsilon} \rceil$, $s > \lceil \frac{3}{\varepsilon^3} \rceil$, and $\delta < 1 - \sqrt{R + 2\varepsilon}$, we have*

$$\mathbf{err}_{\mathrm{comb}}(t, R, s, n, L, \varepsilon, q) < \frac{t}{q} \cdot \left(\lceil \frac{1-R}{\varepsilon} \rceil^2 (1-R) n + \frac{1}{\varepsilon}\right). \tag{8}$$

3. *When $L = \lceil \frac{1}{\varepsilon} \rceil$, $s > \lceil \frac{3}{\varepsilon^2} \rceil$, and $\delta < 1 - \sqrt{R + 2\varepsilon}$, we have*

$$\mathbf{err}_{\mathrm{comb}}(t, R, s, n, L, \varepsilon, q) < \frac{t}{q} \cdot \left(\lceil \frac{1}{\varepsilon} \rceil^2 (1-R) n + \frac{1}{\varepsilon}\right). \tag{9}$$

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A Proof of Corollary 3

Proof. We use induction to prove the result. When $t = 1$, the problem is reduced to Theorem 3. Suppose the result holds for $t - 1$ codewords. For convenience, denote $\langle \alpha_1, \ldots, \alpha_i \rangle \in \mathbb{F}_q^i$ as $\mathbf{z}_i$, $i = 1, \ldots, t$. For $1 \leq i \leq t$, we have

$$\begin{aligned}\mathrm{Bad}(\pi_0, \ldots, \pi_i, \delta) = \{\mathbf{z}_i \in \mathbb{F}_q^i \,|\exists c \in C, \text{ such that } \delta(\pi_0 + \cdots + \alpha_i \pi_i, c) \leq \delta \\ \text{and } \mathrm{Agree}(\pi_0 + \cdots + \alpha_i \pi_i, c) \notin \mathcal{A}_{\delta, \{\pi_0, \ldots, \pi_i\}, C}\}.\end{aligned}$$

For convenience, denote by $A_i, i = 1, \ldots, t$ the case that $\mathbf{z}_t \in \mathrm{Bad}(\pi_0, \ldots, \pi_i, \delta)$. Then we have

$$\begin{aligned}&\Pr_{\mathbf{z}_t \in \mathbb{F}_q^t}(A_t) \\ =&\Pr_{\mathbf{z}_t \in \mathbb{F}_q^t}(A_t \mid A_{t-1}) \cdot \Pr_{\mathbf{z}_{t-1} \in \mathbb{F}_q^{t-1}}(A_{t-1}) + \Pr_{\mathbf{z}_t \in \mathbb{F}_q^t}(A_t \mid \overline{A_{t-1}}) \cdot \Pr_{\mathbf{z}_{t-1} \in \mathbb{F}_q^{t-1}}(\overline{A_{t-1}}) \\ \leq&\Pr_{\mathbf{z}_{t-1} \in \mathbb{F}_q^{t-1}}(A_{t-1}) + \Pr_{\mathbf{z}_t \in \mathbb{F}_q^t}(A_t \mid \overline{A_{t-1}}). \end{aligned} \tag{10}$$

According to our assumption, the first item satisfies

$$\Pr_{\mathbf{z}_{t-1} \in \mathbb{F}_q^{t-1}}(A_{t-1}) < \frac{t-1}{q} \cdot \left(L^2 pn + \frac{1}{\varepsilon}\right). \tag{11}$$

For the second item, for any fixed $\mathbf{z}_{t-1} \notin \mathrm{Bad}(\pi_0, \ldots, \pi_{t-1}, \delta)$, define the set

$$\mathrm{Bad}_{\mathbf{z}_{t-1}} := \{\alpha_t \in \mathbb{F}_q : \langle \mathbf{z}_{t-1}, \alpha_t \rangle \in \mathrm{Bad}(\pi_0, \ldots, \pi_t, \delta)\}.$$

For convenience, define $\pi' := \pi_0 + \cdots + \alpha_{t-1}\pi_{t-1}$. Consider the set

$$\begin{aligned}\mathrm{Bad}(\pi', \pi_t, \delta) = \{\alpha_t \in \mathbb{F}_q \,|\exists c \in C, \text{ such that } \delta(\pi' + \alpha_t \pi_t, c) \leq \delta \\ \text{and } \mathrm{Agree}(\pi' + \alpha_t \pi_t, c) \notin \mathcal{A}_{\delta, \{\pi', \pi_t\}, C}\}.\end{aligned}$$

We want to prove

$$\mathrm{Bad}_{\mathbf{z}_{t-1}} \subseteq \mathrm{Bad}(\pi', \pi_t, \delta). \tag{12}$$

Notice that $|\mathrm{Bad}(\pi', \pi_t, \delta)| < L^2 pn + \frac{1}{\varepsilon}$ by Theorem 3. So if (12) holds, we have

$$|\mathrm{Bad}_{\mathbf{z}_{t-1}}| < L^2 pn + \frac{1}{\varepsilon}. \tag{13}$$

Suppose $\mathcal{A}_{\delta, \{\pi', \pi_t\}, C} = \{D'_1, \ldots, D'_{m'}\}$. Since we have $\mathbf{z}_{t-1} \notin \mathrm{Bad}(\pi_0, \ldots, \pi_{t-1}, \delta)$, $\{D'_1, \ldots, D'_{m'}\}$ are δ-correlated-agree-domains of $\pi_0, \cdots, \pi_{t-1}$ (may be not maximal). Otherwise, suppose, without loss of generality, D'_1 is not a δ-correlated-agree-domain between $\{\pi_0, \ldots, \pi_{t-1}\}$ and C, then $\mathbf{z}_{t-1}$ expands the correlated-agree-domain. This is a contradiction to $\mathbf{z}_{t-1} \notin \mathrm{Bad}(\pi_0, \ldots, \pi_{t-1}, \delta)$.

Thus, $\{D'_1, \ldots, D'_{m'}\}$ are δ-correlated-agree-domains of $\pi_0, \cdots, \pi_t$ according to the definition of $\mathcal{A}_{\delta, \{\pi', \pi_t\}, C}$. As a result, for any $D' \in \mathcal{A}_{\delta, \{\pi', \pi_t\}, C}$, $\exists D \in \mathcal{A}_{\delta, \{\pi_0, \ldots, \pi_t\}, C}$, such that $D' \subseteq D$. On the other hand, D' is maximal when considering π' and π_t according to the definition of $\mathcal{A}_{\delta, \{\pi', \pi_t\}, C}$. Then we have $D' = D$ since D is a correlated-agreement between π', π_t and C. Thus,

$$\mathcal{A}_{\delta, \{\pi', \pi_t\}, C} \subseteq \mathcal{A}_{\delta, \{\pi_0, \ldots, \pi_t\}, C}. \tag{14}$$

Fix an $\alpha \in \mathrm{Bad}_{\mathbf{z}_{t-1}}$, we have $\langle \mathbf{z}_{t-1}, \alpha \rangle \in \mathrm{Bad}(\pi_0, \ldots, \pi_t, \delta)$. According to the definition of $\mathrm{Bad}_\delta(\pi_0, \ldots, \pi_t)$, $\exists c \in C$ such that

$$\mathrm{Agree}(\pi' + \alpha\pi_t, c) \notin \mathcal{A}_{\delta,\{\pi_0,\ldots,\pi_t\},C} \text{ and } |\mathrm{Agree}(\pi' + \alpha\pi_t, c)| \geq (1-\delta)n.$$

(14) tells us $\mathrm{Agree}(\pi' + \alpha\pi_t, c) \notin \mathcal{A}_{\delta,\{\pi',\pi_t\},C}$. Thus,

$$\alpha \in \mathrm{Bad}(\pi', \pi_t, \delta)$$

and we have proved (12).

Plugging (11) and (13) into (10), we have

$$\mathrm{Pr}_{\mathbf{z}_t \in \mathbb{F}_q^t}(A_t) < \frac{t}{q} \cdot \left(L^2 pn + \frac{1}{\varepsilon} \right).$$